In [1]:
```
import numpy as np
```

In [2]:
```
ones_arr = np.ones((5,5),dtype=int)
```

In [3]:
```
ones_arr
```

Out[3]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [4]:
```
ones_arr * 255
```

Out[4]:
```
array([[255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255]])
```

In [5]:
```
import matplotlib.pyplot as plt
```

In [6]:
```
#!matplotlib inline # all the graph should keep inside the line
```

In [7]:
```
from PIL import Image # python imaging library
```

In [10]:
```
eye = Image.open(r'C:\Users\Lenovo\Downloads\image1.jpg')
```

In [11]:
```
eye
```

Out[11]:

In [12]:
```
eye_img = Image.open(r'C:\Users\Lenovo\Downloads\image1.jpg')
```

In [13]:
```
eye_img
```

Out[13]:

In [14]:
```
type(eye_img)
```

Out[14]:
```
PIL.JpegImagePlugin.JpegImageFile
```

In [15]:
```
eye_arr = np.asarray(eye_img)
eye_arr
```

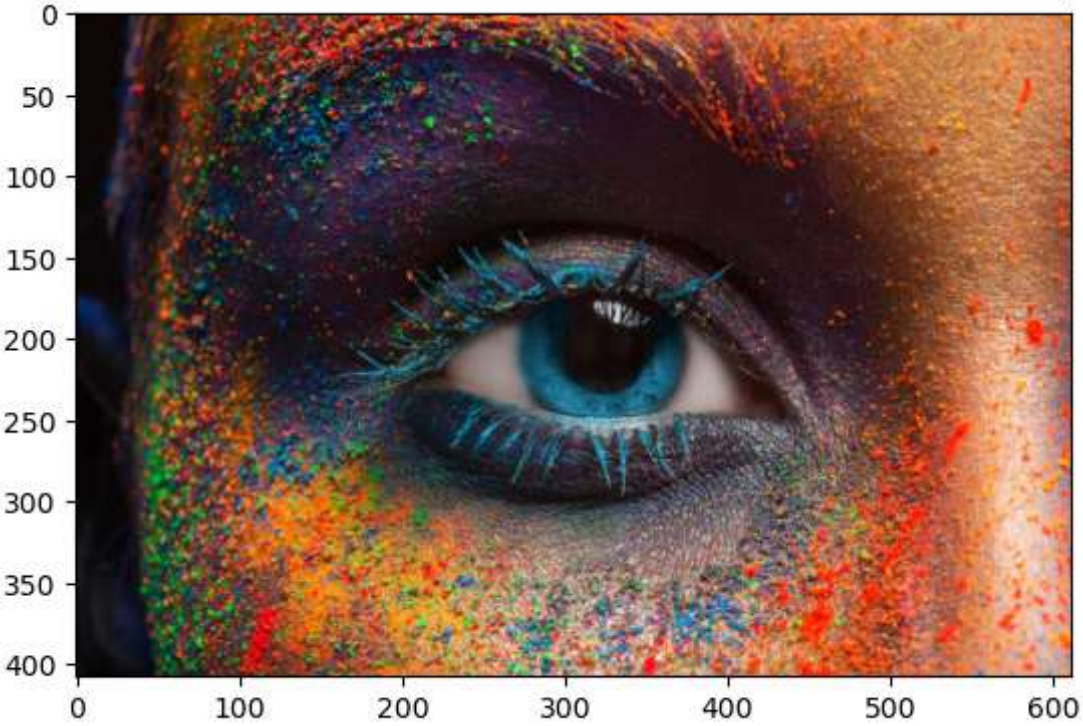
Out[15]:
```
array([[[ 12,   6,   6],
        [ 12,   6,   6],
        [ 12,   6,   6],
        ...,
        [226, 141, 121],
        [226, 141, 112],
        [211, 123,  85]],

       [[ 12,   6,   6],
        [ 12,   6,   6],
        [ 12,   6,   6],
        ...,
        [231, 138,  81],
        [233, 139,  77],
        [225, 132,  65]],

       [[ 12,   6,   6],
        [ 12,   6,   6],
        [ 12,   6,   6],
        ...,
        [247, 141,  41],
        [249, 143,  43],
        [247, 144,  43]],

       ...,

       [[ 19,  18,  24],
        [ 18,  17,  23],
        [ 16,  15,  21],
        ...,
        [253, 158, 100],
        [239, 147, 106],
        [223, 135, 111]],

       [[ 19,  18,  24],
        [ 18,  17,  23],
        [ 17,  16,  22],
        ...,
        [255, 151,  94],
        [240, 142, 107],
        [221, 129, 114]],

       [[ 19,  18,  24],
        [ 18,  17,  23],
        [ 17,  16,  22],
        ...,
        [252, 148,  95],
        [237, 142, 114],
        [219, 131, 127]]], dtype=uint8)
```

In [16]:
```
type(eye_arr)
```

Out[16]:
```
numpy.ndarray
```

In [17]:
```
plt.imshow(eye_arr)
#plt.show(eye_arr) new os
```

Out[17]:
```
<matplotlib.image.AxesImage at 0x1f85fd8cc80>
```



In [18]:
```
eye_arr.shape
```

Out[18]: (408, 612, 3)

In [19]:
```
eye_red = eye_arr.copy()
```

In [20]:
```
eye_red
```

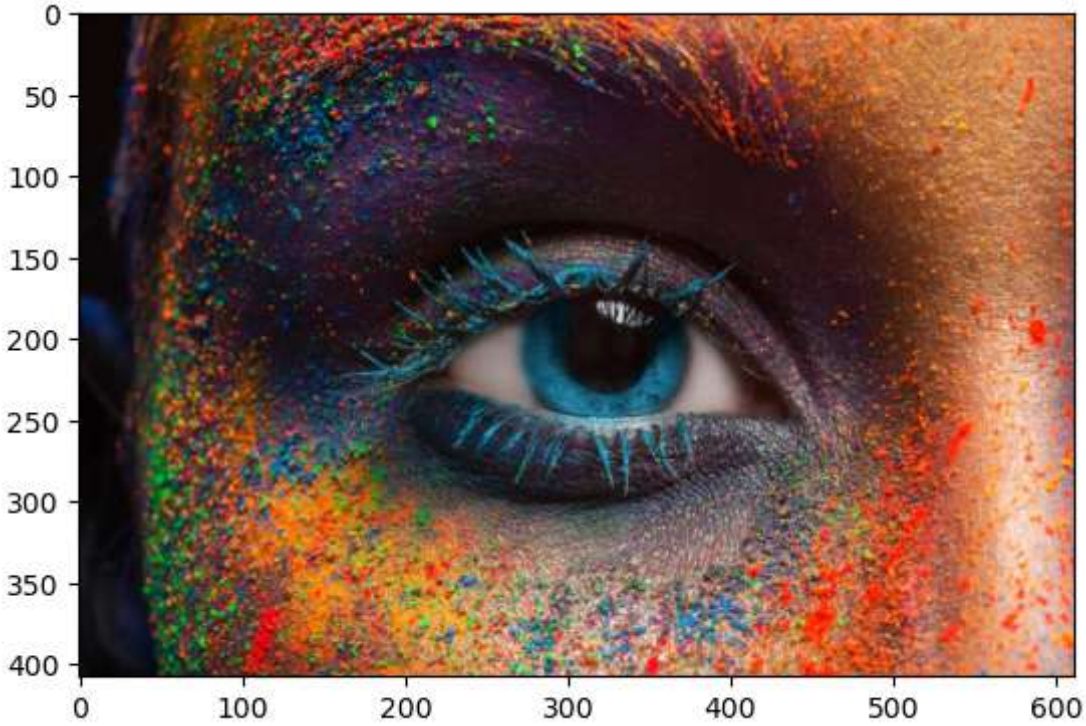
Out[20]:
```
array([[[ 12,   6,   6],
        [ 12,   6,   6],
        [ 12,   6,   6],
        ...,
        [226, 141, 121],
        [226, 141, 112],
        [211, 123,  85]],

       [[ 12,   6,   6],
        [ 12,   6,   6],
        [ 12,   6,   6],
        ...,
        [231, 138,  81],
        [233, 139,  77],
        [225, 132,  65]],

       [[ 12,   6,   6],
        [ 12,   6,   6],
        [ 12,   6,   6],
        ...,
        [247, 141,  41],
        [249, 143,  43],
        [247, 144,  43]],

       ...,

       [[ 19,  18,  24],
        [ 18,  17,  23],
        [ 16,  15,  21],
        ...,
        [253, 158, 100],
        [239, 147, 106],
        [223, 135, 111]],

       [[ 19,  18,  24],
        [ 18,  17,  23],
        [ 17,  16,  22],
        ...,
        [255, 151,  94],
        [240, 142, 107],
        [221, 129, 114]],

       [[ 19,  18,  24],
        [ 18,  17,  23],
        [ 17,  16,  22],
        ...,
        [252, 148,  95],
        [237, 142, 114],
        [219, 131, 127]]], dtype=uint8)
```

In [21]:
```
eye_arr == eye_red
```

Out[21]:
```
array([[[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       ...,

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]]])
```

In [22]:
```
plt.imshow(eye_red)
```

Out[22]:
```
<matplotlib.image.AxesImage at 0x1f861f19820>
```

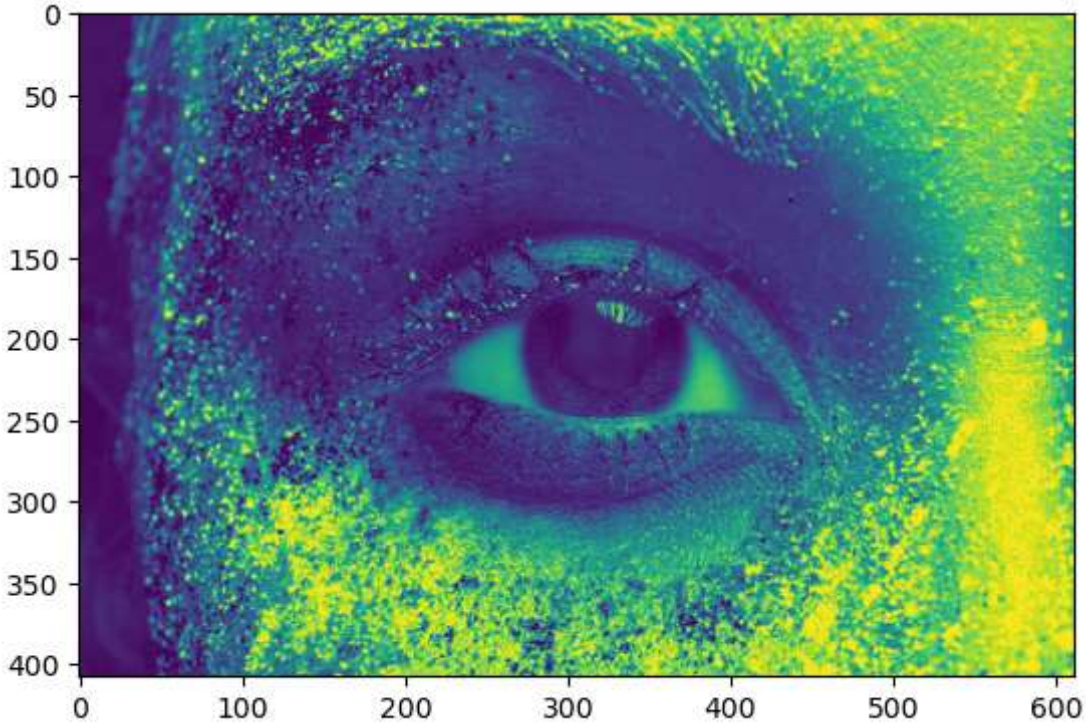


In [23]:
```
eye_red.shape
```

Out[23]: (408, 612, 3)

In [24]:
```
# R G B
plt.imshow(eye_red[:,:,0])
```

Out[24]: <matplotlib.image.AxesImage at 0x1f861f2a2a0>

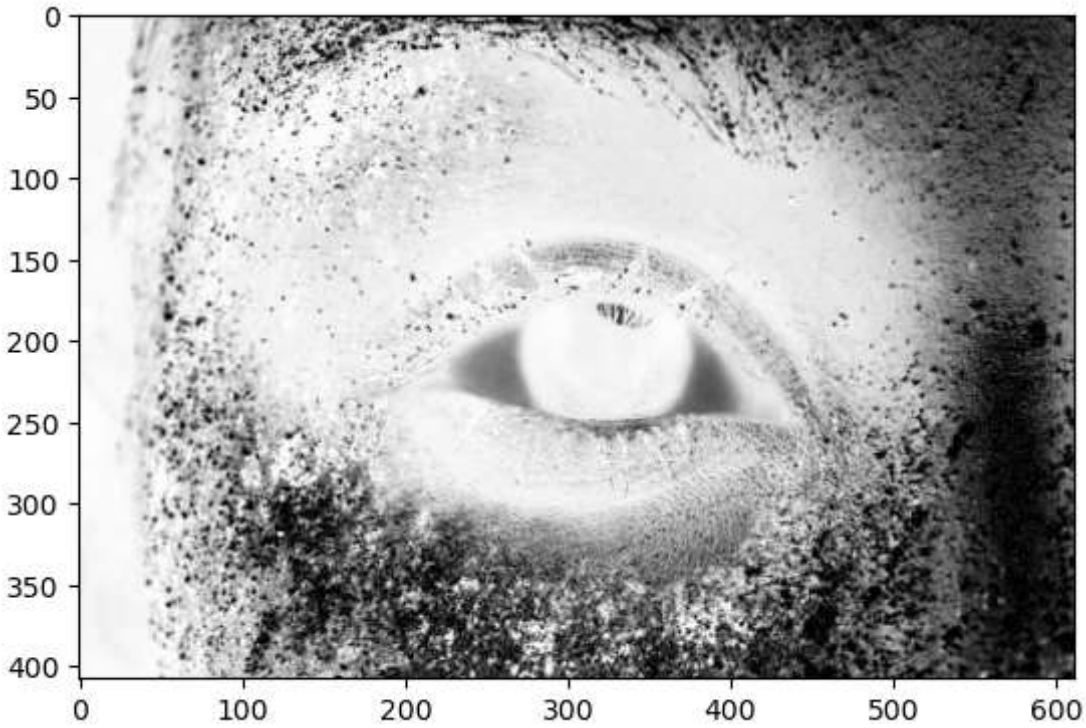


In [25]:
```
eye_red[:,:,0]
```

Out[25]:
```
array([[ 12,  12,  12, ..., 226, 226, 211],
       [ 12,  12,  12, ..., 231, 233, 225],
       [ 12,  12,  12, ..., 247, 249, 247],
       ...,
       [ 19,  18,  16, ..., 253, 239, 223],
       [ 19,  18,  17, ..., 255, 240, 221],
       [ 19,  18,  17, ..., 252, 237, 219]], dtype=uint8)
```

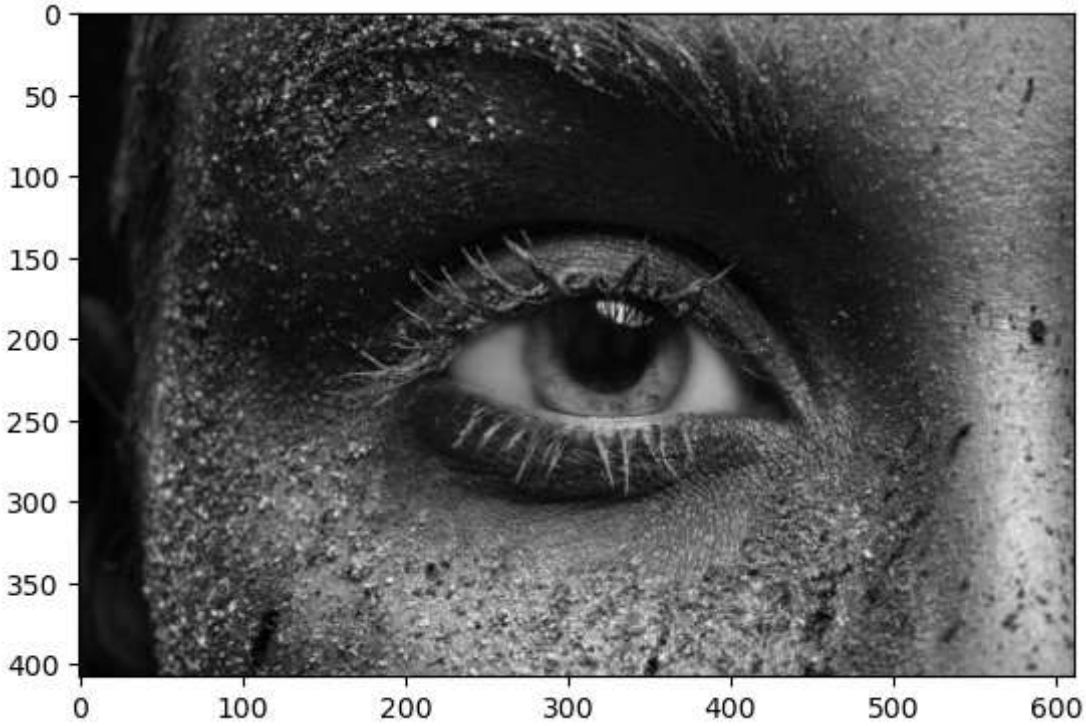
In [26]:
```
plt.imshow(eye_red[:,:,0], cmap='Greys')
```

Out[26]:
```
<matplotlib.image.AxesImage at 0x1f861f19a60>
```



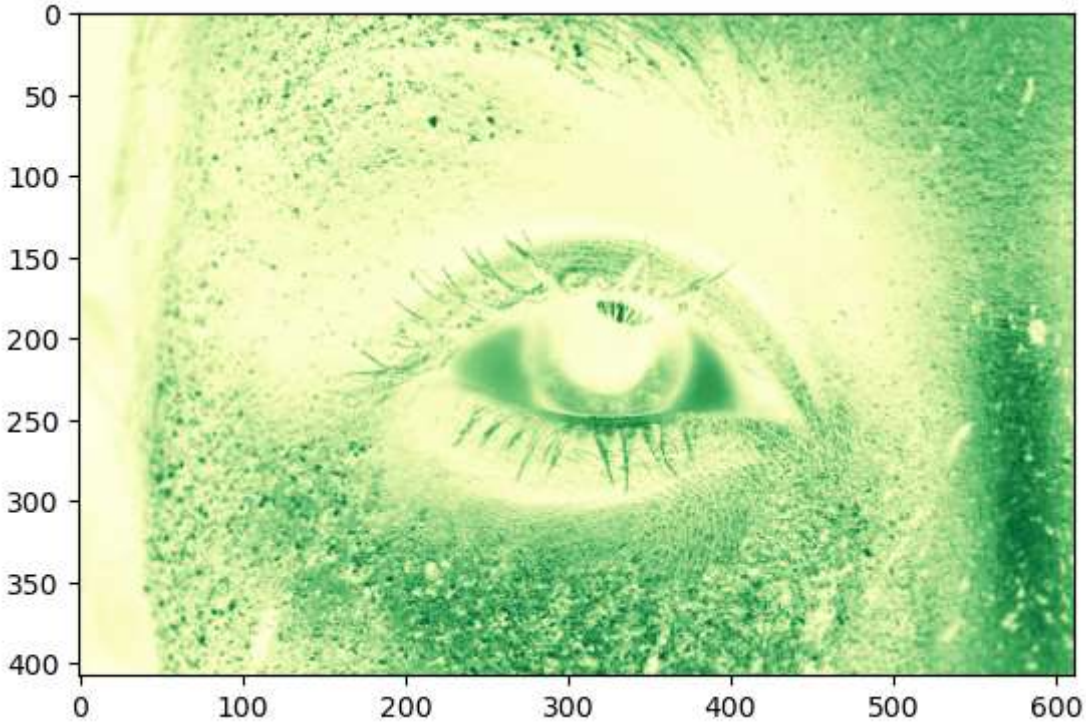
In [27]:
```
plt.imshow(eye_red[:,:,1], cmap='grey')
```

Out[27]:
```
<matplotlib.image.AxesImage at 0x1f861f17f80>
```



In [28]:
```
plt.imshow(eye_red[:,:,1], cmap='YlGn')
#plt.show()
```

Out[28]: <matplotlib.image.AxesImage at 0x1f864391880>



In [29]:
```
eye_red[:,:,0]
```

Out[29]:
```
array([[ 12,  12,  12, ..., 226, 226, 211],
       [ 12,  12,  12, ..., 231, 233, 225],
       [ 12,  12,  12, ..., 247, 249, 247],
       ...,
       [ 19,  18,  16, ..., 253, 239, 223],
       [ 19,  18,  17, ..., 255, 240, 221],
       [ 19,  18,  17, ..., 252, 237, 219]], dtype=uint8)
```

In [30]:
```
eye_red[:,:,1]
```

Out[30]:
```
array([[  6,   6,   6, ..., 141, 141, 123],
       [  6,   6,   6, ..., 138, 139, 132],
       [  6,   6,   6, ..., 141, 143, 144],
       ...,
       [ 18,  17,  15, ..., 158, 147, 135],
       [ 18,  17,  16, ..., 151, 142, 129],
       [ 18,  17,  16, ..., 148, 142, 131]], dtype=uint8)
```

In [31]:
```
eye_red[:,:,2]
```

Out[31]:
```
array([[  6,   6,   6, ..., 121, 112,  85],
       [  6,   6,   6, ...,  81,  77,  65],
       [  6,   6,   6, ...,  41,  43,  43],
       ...,
       [ 24,  23,  21, ..., 100, 106, 111],
       [ 24,  23,  22, ...,  94, 107, 114],
       [ 24,  23,  22, ...,  95, 114, 127]], dtype=uint8)
```

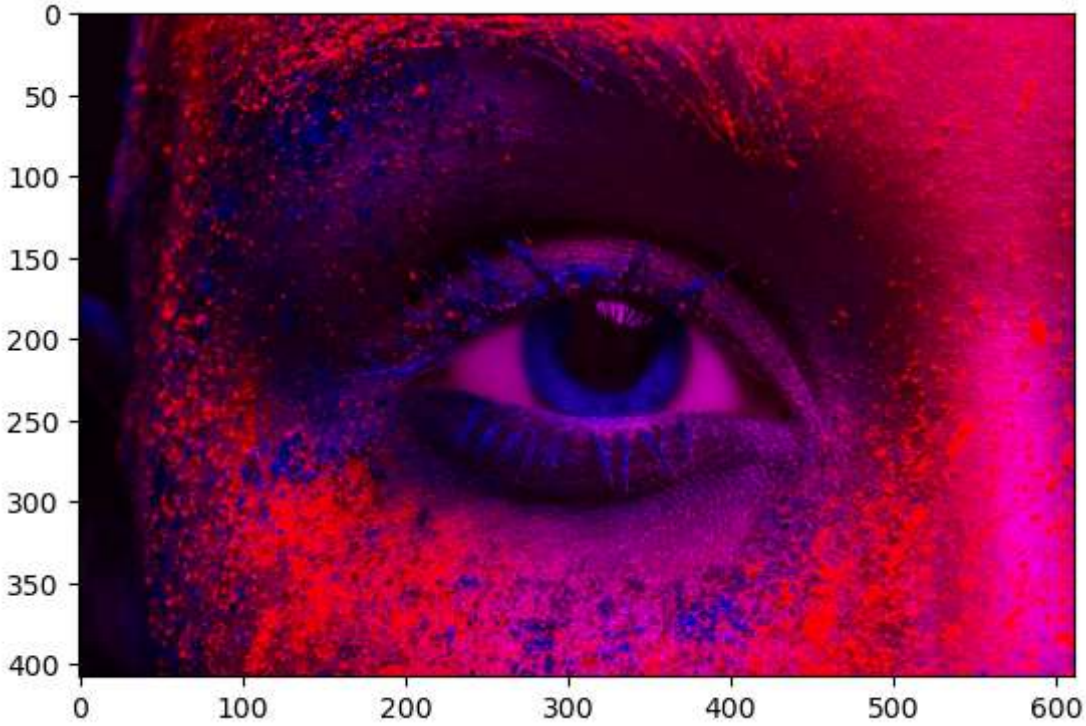
In [32]:
```
eye_red[:,:,1] = 0
```

In [33]:
```
eye_red[:,:,1]
```

Out[33]:
```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]], dtype=uint8)
```

In [34]:
```
plt.imshow(eye_red)
```

Out[34]:
```
<matplotlib.image.AxesImage at 0x1f864409820>
```



In [35]:
```
eye_red[:,:,2]
```

Out[35]:
```
array([[  6,   6,   6, ..., 121, 112,  85],
       [  6,   6,   6, ...,  81,  77,  65],
       [  6,   6,   6, ...,  41,  43,  43],
       ...,
       [ 24,  23,  21, ..., 100, 106, 111],
       [ 24,  23,  22, ...,  94, 107, 114],
       [ 24,  23,  22, ...,  95, 114, 127]], dtype=uint8)
```

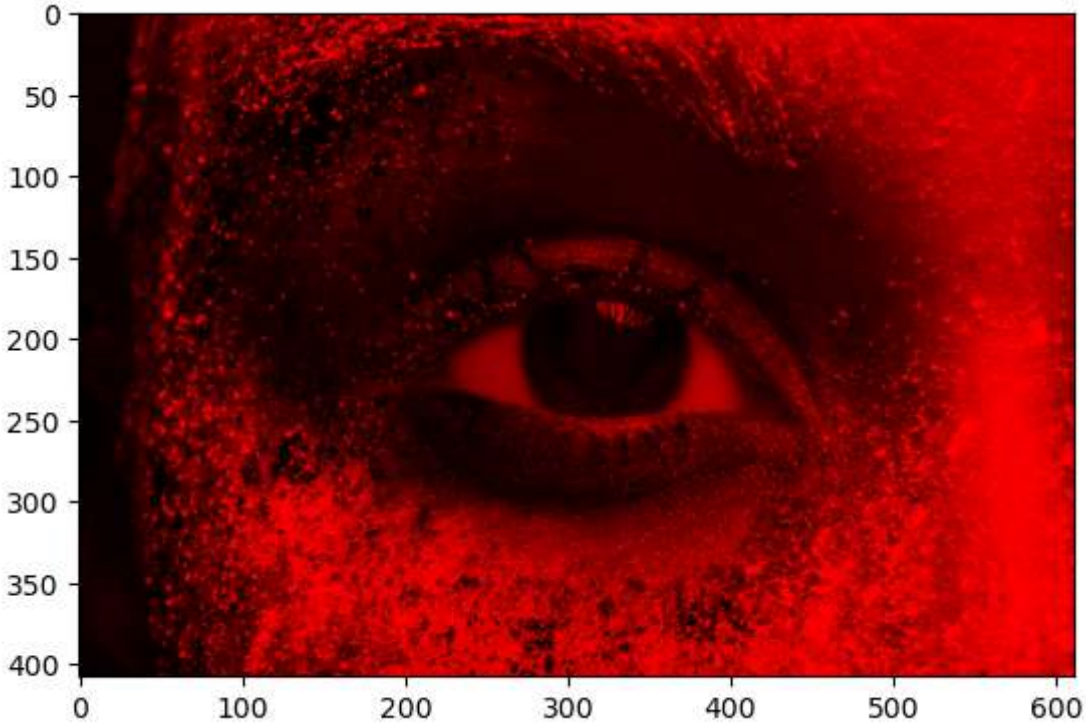
In [36]:
```
eye_red[:,:,2]  = 0
```

In [37]:
```
eye_red[:,:,2]
```

Out[37]:
```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]], dtype=uint8)
```

In [38]:
```
plt.imshow(eye_red)
```

Out[38]:
```
<matplotlib.image.AxesImage at 0x1f862ad22a0>
```



In [39]:
```
eye_arr
```

Out[39]:
```
array([[[ 12,   6,   6],
        [ 12,   6,   6],
        [ 12,   6,   6],
        ...,
        [226, 141, 121],
        [226, 141, 112],
        [211, 123,  85]],

       [[ 12,   6,   6],
        [ 12,   6,   6],
        [ 12,   6,   6],
        ...,
        [231, 138,  81],
        [233, 139,  77],
        [225, 132,  65]],

       [[ 12,   6,   6],
        [ 12,   6,   6],
        [ 12,   6,   6],
        ...,
        [247, 141,  41],
        [249, 143,  43],
        [247, 144,  43]],

       ...,

       [[ 19,  18,  24],
        [ 18,  17,  23],
        [ 16,  15,  21],
        ...,
        [253, 158, 100],
        [239, 147, 106],
        [223, 135, 111]],

       [[ 19,  18,  24],
        [ 18,  17,  23],
        [ 17,  16,  22],
        ...,
        [255, 151,  94],
        [240, 142, 107],
        [221, 129, 114]],

       [[ 19,  18,  24],
        [ 18,  17,  23],
        [ 17,  16,  22],
        ...,
        [252, 148,  95],
        [237, 142, 114],
        [219, 131, 127]]], dtype=uint8)
```

In [40]:
```
eye_red
```

Out[40]:
```
array([[[ 12,   0,   0],
        [ 12,   0,   0],
        [ 12,   0,   0],
        ...,
        [226,   0,   0],
        [226,   0,   0],
        [211,   0,   0]],

       [[ 12,   0,   0],
        [ 12,   0,   0],
        [ 12,   0,   0],
        ...,
        [231,   0,   0],
        [233,   0,   0],
        [225,   0,   0]],

       [[ 12,   0,   0],
        [ 12,   0,   0],
        [ 12,   0,   0],
        ...,
        [247,   0,   0],
        [249,   0,   0],
        [247,   0,   0]],

       ...,

       [[ 19,   0,   0],
        [ 18,   0,   0],
        [ 16,   0,   0],
        ...,
        [253,   0,   0],
        [239,   0,   0],
        [223,   0,   0]],

       [[ 19,   0,   0],
        [ 18,   0,   0],
        [ 17,   0,   0],
        ...,
        [255,   0,   0],
        [240,   0,   0],
        [221,   0,   0]],

       [[ 19,   0,   0],
        [ 18,   0,   0],
        [ 17,   0,   0],
        ...,
        [252,   0,   0],
        [237,   0,   0],
        [219,   0,   0]]], dtype=uint8)
```

In [41]:
```
eye_img
```

Out[41]:

In [42]:
```
arr1 = np.asarray(eye_img)
```

In [43]:
```
type(arr1)
```

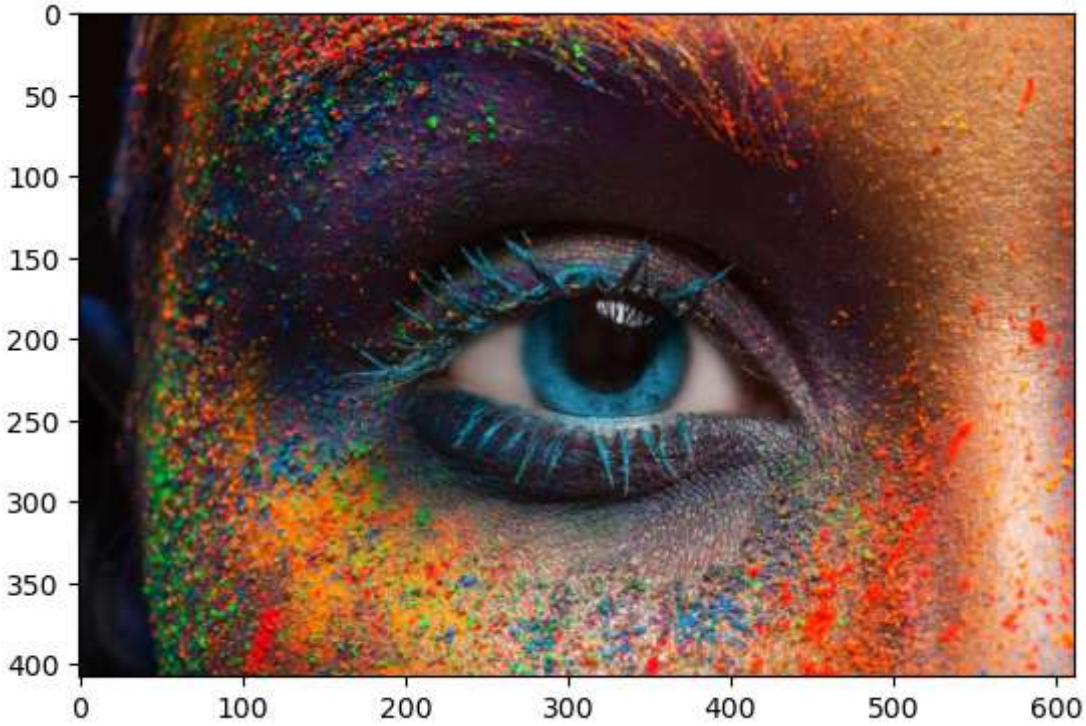
Out[43]:
```
numpy.ndarray
```

In [44]:
```
arr1.shape
```

Out[44]:
```
(408, 612, 3)
```

In [45]:
```
plt.imshow(arr1)
```

Out[45]:
```
<matplotlib.image.AxesImage at 0x1f862b1fd10>
```

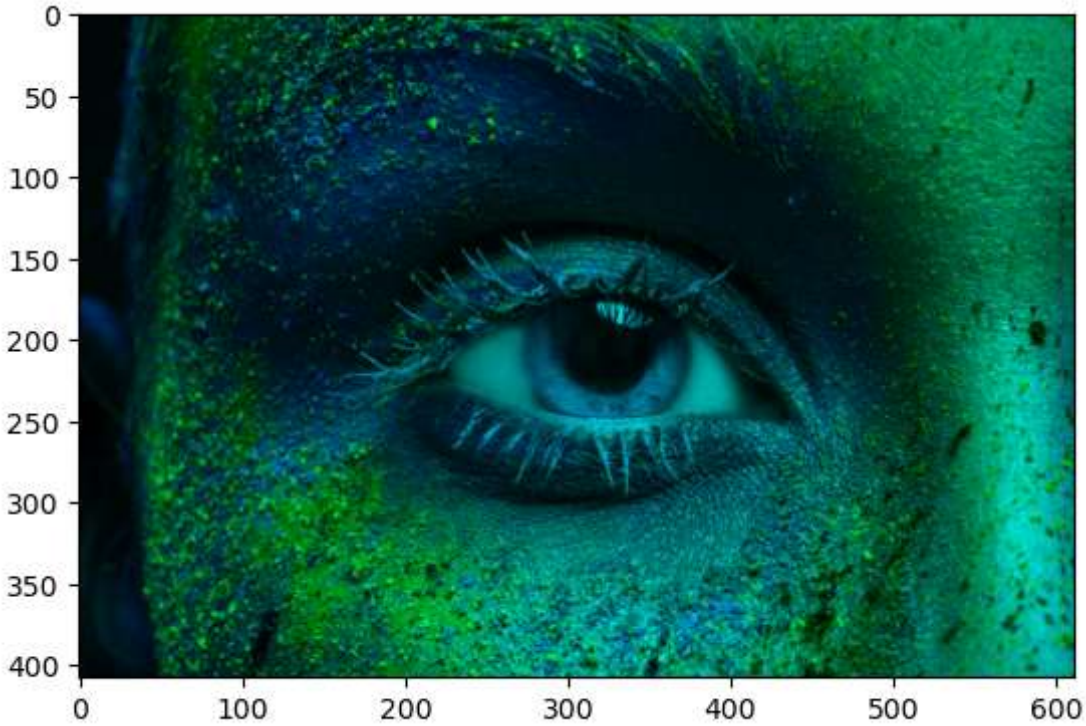


In [46]:
```
eye_img1 = arr1.copy()
```

In [48]:
```
eye_img1[:,:,0] = 0
```

In [49]:
```
plt.imshow(eye_img1)
```

Out[49]:
```
<matplotlib.image.AxesImage at 0x1f864442120>
```



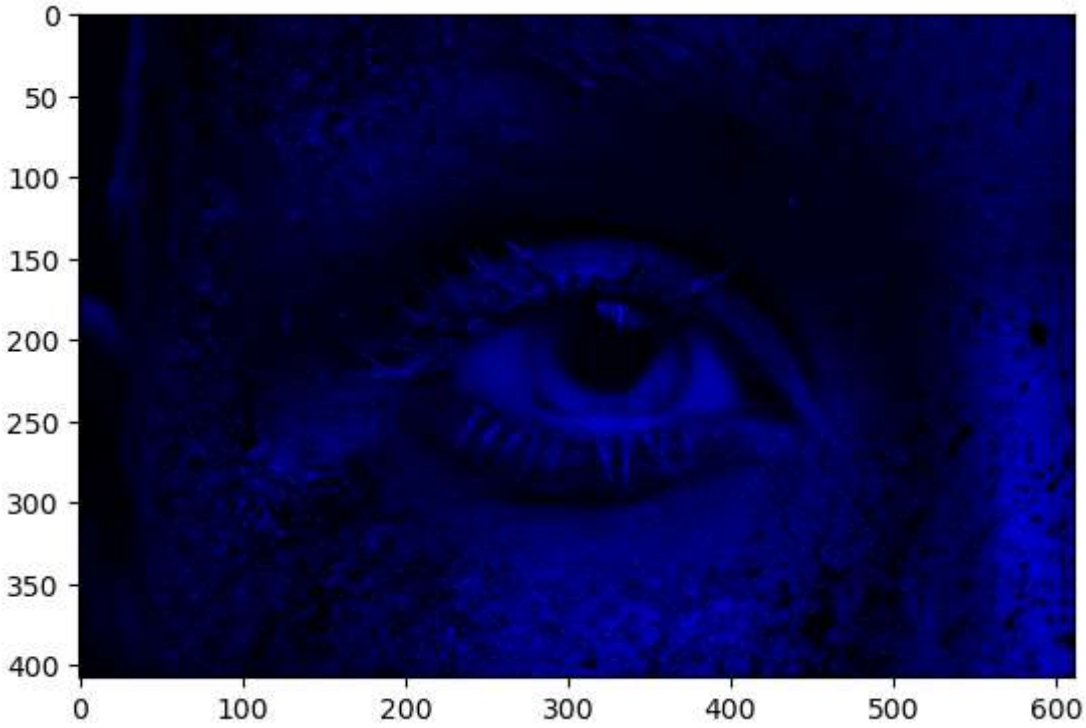
In [50]:
```
eye_img1[:,:,1]
```

Out[50]:
```
array([[  6,   6,   6, ..., 141, 141, 123],
       [  6,   6,   6, ..., 138, 139, 132],
       [  6,   6,   6, ..., 141, 143, 144],
       ...,
       [ 18,  17,  15, ..., 158, 147, 135],
       [ 18,  17,  16, ..., 151, 142, 129],
       [ 18,  17,  16, ..., 148, 142, 131]], dtype=uint8)
```

In [51]:
```
eye_img1[:,:,1] = 0
```

In [52]:
```
plt.imshow(eye_img1)
```

Out[52]:
```
<matplotlib.image.AxesImage at 0x1f8646a5df0>
```



In [ ]: